Test Plan for NumberToWords Conversion API

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1. Objective

This document outlines the test plan for the NumberToWords Conversion API, which returns the word representation of a

positive number passed as a parameter. The objective is to ensure that all functionalities work correctly, are free of

defects, and provide accurate conversions within the allowed number range (limited to quadrillions).

2. Scope

Features to be tested:

- Conversion of numeric values to words.
- Handling of valid and invalid inputs.
- API response format and structure.
- Performance and response time.
- Security measures such as unauthorized access handling.

Types of testing:

- Functional Testing
- Boundary Value Analysis
- Performance Testing
- Negative Testing
- Regression Testing
- Security Testing

Environments:

- Operating Systems: Windows 10, macOS, Linux.

- Browsers (if applicable): Google Chrome, Mozilla Firefox, Microsoft Edge.
- Devices: Desktop computers, laptops, tablets, smartphones.
- Network Connectivity: Wi-Fi, cellular, wired connections.
- API Testing Tools: Postman, SoapUI.
- Security Protocols: Authentication tokens, certificates, SSL/TLS.

Evaluation Criteria:

- Number of defects identified.
- API response time.
- Test case execution coverage.
- User satisfaction with API accuracy.

Team roles and responsibilities:

- Test Lead: Oversees the testing process, assigns tasks.
- Testers: Execute test cases and report defects.
- Developers: Fix reported issues and improve API.
- Project Manager: Ensures timely delivery of API testing.

3. Test Strategy

Step 1: Test Scenarios and Test Cases Creation

- Equivalence Class Partitioning.
- Boundary Value Analysis.
- Error Guessing.

Step 2: Testing Procedure

- Smoke Testing: Verify basic functionality.
- In-depth Testing: Execute all test cases once smoke tests pass.
- Defect Reporting: Report defects in the tracking tool.
- Regression Testing: Ensure new changes do not break existing functionality.

Step 3: Best Practices

- Shift Left Testing: Test early in the development cycle.
- Exploratory Testing: Identify issues beyond predefined test cases.
- End-to-End Flow Testing: Ensure the API works as expected in different scenarios.

4. Test Schedule

- Test Planning: 1 week
- Test Case Creation: 1 week
- Test Execution: 2 weeks
- Defect Reporting & Fixing: Ongoing during execution
- Test Summary Report Submission: 1 week

5. Test Deliverables

- Test Plan Document
- Test Scenarios and Test Cases
- Test Execution Report
- Defect Reports
- Test Summary Report

6. Entry and Exit Criteria

Requirement Analysis:

- Entry: Received requirement specifications.
- Exit: Documented and reviewed test scenarios.

Test Execution:

- Entry: API build ready, test cases approved.
- Exit: All test cases executed, defect triage completed.

Test Closure:

- Entry: Test execution completed, all critical defects resolved.

- Exit: Final test summary report submitted.

7. Tools

- Defect Tracking: JIRA, Bugzilla.
- Test Execution: Postman, SoapUI.
- Performance Testing: JMeter.
- Documentation: MS Word, Excel, Confluence.

8. Risks and Mitigations

Possible Risks:

- Non-availability of API for testing.
- Unstable API responses leading to inconsistent results.
- Limited time for testing.

Mitigation Strategies:

- Ensure backup test environments.
- Report stability issues early.
- Prioritize critical test cases if time is limited.

9. Approvals

Documents for Client Approval:

- Test Plan
- Test Scenarios & Test Cases
- Test Execution Reports
- Defect Reports
- Final Test Summary Report